

The spectrum of limit models in stable AECs

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Based largely on joint work with Marcos Mazari-Armida

Overview

1. Background - AECs
2. Background - Limit models
3. Papers 1 ([BeMa]) and 2 ([Bea25b]) - The spectrum of limit models
4. Paper 3 ([Bea25a]) - Extending to 'weak' independence relations
5. Paper 4 ([Bea26]) - Disjoint non-forking amalgamation

Background - Setting the scene

I suppose we do model theory because we are interested broadly in abstractly studying classes of mathematical structures. One broad theme of interest is *classification theory*, i.e. drawing 'dividing lines' between different classes, which normally separate 'well behaved' classes (e.g. superstable, stable, NIP, categorical) from 'chaotic' classes.

In first order model theory, the magic 'elementary substructure' relation $M \preceq N$ is very well behaved. We have lots of handy tools like compactness, nicely behaved chains, the Löwenheim Skolem theorem, and so on.

But this framework is sometimes a little restrictive - there are certain classes we cannot axiomatise, and sometimes we care about different substructure relations than $\preceq$.

Background - Where are we heading...

One answer to this is *abstract elementary classes (AECs)*, which provide a framework that generalises lots of this first order machinery to a broader setting.

AECs are essentially classes of models K with a substructure relation $\leq_K$ that satisfies some of the nice properties of the elementary substructure relation $\preceq$ on models of a first order theory T . This provides a more general setting in which we can attempt classification theory.

However, we lose compactness, and formulas are less naturally useful.

Background - ACs

Definition

$\mathbf{K} = (K, \leq_{\mathbf{K}})$ is an abstract class (AC) provided that

1. K is a class of models in a language $L(\mathbf{K})$ closed under isomorphisms
2. $\leq_{\mathbf{K}}$ is a relation on K which respects isomorphisms and strengthens the substructure relation $\subseteq$

Notation

Given a cardinal λ , $K_{\lambda} = \{M \in \mathbf{K} : \|M\| = \lambda\}$, and $\mathbf{K}_{\lambda} = (K_{\lambda}, \leq_{\mathbf{K}})$

Background - AECs

Definition (Shelah)

An abstract class $\mathbf{K} = (K, \leq_{\mathbf{K}})$ is an *abstract elementary class (AEC)* provided that it satisfies:

1. (Coherence) If $M_1 \subseteq M_2$, $M_1 \leq_{\mathbf{K}} M_3$ and $M_2 \leq_{\mathbf{K}} M_3$, then $M_1 \leq_{\mathbf{K}} M_2$
2. (Tarski-Vaught chain axiom) For all $\leq_{\mathbf{K}}$ -increasing chains $\langle M_i : i < \delta \rangle$
 - $M = \bigcup_{i < \delta} M_i \in K$ and $M_i \leq_{\mathbf{K}} M$ for all $i < \delta$
 - if $M_i \leq_{\mathbf{K}} N$ for all $i < \delta$, $M \leq_{\mathbf{K}} N$
3. (Löwenheim Skolem axiom) there is a cardinal $\text{LS}(\mathbf{K})$ such that for all $N \in \mathbf{K}$ and $A \subseteq |N|$, there exists $M \leq_{\mathbf{K}} N$ such that $A \subseteq |M|$ and $\|M\| \leq |A| + \text{LS}(\mathbf{K})$

When T is a first order theory, $(\text{Mod}(T), \preceq)$ satisfies these hypotheses with $\text{LS}(\mathbf{K}) = |L(T)| + \aleph_0$, but the framework is much more general.

Background - Examples of AECs

Examples

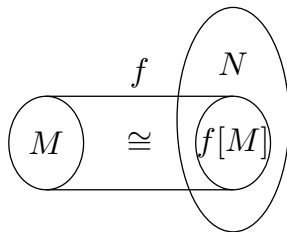
1. $\mathbf{K} = (\text{Mod}(T), \preceq)$ for a first order theory T , with $\text{LS}(\mathbf{K}) = |\text{L}(T)| + \aleph_0$
2. $\mathbf{K} = (\{M \models T : M \text{ is } \aleph_0\text{-homogeneous and } D(M) = \Gamma\}, \preceq)$ for a first order theory T where $\Gamma \subseteq D(M)$, with $\text{LS}(\mathbf{K}) = |\text{L}(T)| + \aleph_0$
3. $\mathbf{K} = (\text{Mod}(\varphi), \subseteq)$ for a universal sentence $\varphi \in \text{L}_{\infty, \omega}$, with $\text{LS}(\mathbf{K}) = |\text{L}(\varphi)| + \aleph_0$
4. $\mathbf{K} = (\text{Mod}(\varphi), \preceq_F)$ for some sentence $\varphi \in \text{L}_{\infty, \omega}$ and F a fragment containing φ , with $\text{LS}(\mathbf{K}) = |F| + |\text{L}(\varphi)| + \aleph_0$
5. $\mathbf{K} = (\text{Mod}(\varphi), \preceq_F^*)$ for some sentence $\varphi \in \text{L}_{\omega_1, \omega}(Q)$ and F the smallest countable fragment containing φ , with $\text{LS}(\mathbf{K}) = \aleph_1$.
6. To name a few of the many algebraic examples,
 - 6.1 Torsion Abelian groups with $\subseteq$, pure subgroup, or $\preceq$
 - 6.2 Modules over a fixed ring with $\subseteq$, pure submodule, or $\preceq$
 - 6.3 Actions of a fixed monoid with $\subseteq$ or $\preceq$

Background - Some notions which lift

In this setting, we can generalise many familiar notions, such as embeddings (to *strong* or $\mathbf{K}$ -embeddings), types (to *Galois/orbital types*), stability, saturation, and more. While we keep lots of nice machinery, we lose lots of useful tools as well - in particular compactness.

Definition

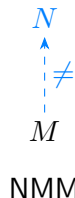
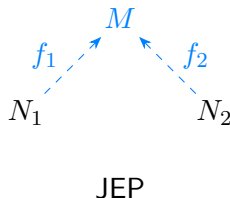
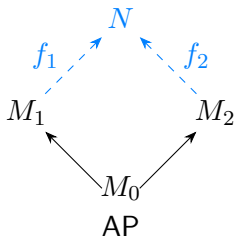
Let $\mathbf{K}$ be an AC. A $\mathbf{K}$ -embedding is an embedding $f : M \rightarrow N$ where $M, N \in \mathbf{K}$ and $f[M] \leq_{\mathbf{K}} N$.



These generalise first order elementary embeddings.

Background - Extra structural assumptions: AP, JEP, NMM

We will sometimes assume further that AC or AEC $\mathbf{K}$ has amalgamation, joint embedding, and no maximal models.



$(\text{Mod}(T), \preceq)$ satisfies these hypotheses if T is complete with infinite models (but again, our context is much broader).

In AECs, AP, JEP, and NMM together are equivalent to existence of monster models. AP in particular makes (Galois) types behave well.

Background - Quick look at Galois/orbital types

Definition

Let $\mathbf{K}$ be an AC satisfying AP. Define a relation $\sim$ on $\{(a, M, N) : M \leq_{\mathbf{K}} N, a \in N\}$ by $(a_1, M_1, N_1) \sim (a_2, M_2, N_2)$ if $M_1 = M_2 =: M$ and there exist N, f_1, f_2 as in the commuting diagram below:

$$\begin{array}{ccc} a_1 \in N_1 & \overset{f_1}{\dashrightarrow} & N \\ \uparrow & & \uparrow f_2 \\ M & \longrightarrow & N_2 \\ & & \downarrow \Psi \\ & & a_2 \end{array} \quad f_1(a_1) = f_2(a_2)$$

$\sim$ is an equivalence relation, so define $\mathbf{gtp}(a/M, N) := [(a, M, N)]_{\sim}$ as the *Galois/orbital type* of a over M in N . The set of Galois types over $M \in \mathbf{K}$ is $\mathbf{gS}(M) = \{\mathbf{gtp}(a/M, N) : M \leq_{\mathbf{K}} N, a \in N\}$.

Background - Extra structural assumptions: Independence relations

Intuitively, an independence relation is a relation on types that should behave similarly to first order forking, which is itself a kind of generalisation of algebraic and vector space independence. When they exist, these relations are a very useful tool.

Definition

Let $\mathbf{K}$ be an AC. An *independence relation* (on $\mathbf{K}$) is a relation $\perp$ on tuples (M_0, a, M, N) where $M_0 \leq_{\mathbf{K}} M \leq_{\mathbf{K}} N$ and $a \in N$ satisfying *invariance* and *monotonicity*. We write $a \underset{M_0}{\overset{N}{\perp}} M$ as a shorthand for $(M_0, a, M, N) \in \perp$.

Here *invariance* says if $a \underset{M_0}{\overset{N}{\perp}} M$ and $f : N \rightarrow N'$ is a $\mathbf{K}$ -embedding, then $f(a) \underset{f[M_0]}{\overset{N'}{\perp}} f[M]$.

Monotonicity says $a \underset{M_0}{\overset{N}{\perp}} M$ and $M_0 \leq_{\mathbf{K}} M_1 \leq_{\mathbf{K}} M_2 \leq_{\mathbf{K}} M \leq_{\mathbf{K}} N_0 \leq_{\mathbf{K}} N$ imply $a \underset{M_1}{\overset{N_0}{\perp}} M_2$.

Background - Independence relations continued

Definition

Suppose $\perp$ an independence relation on $\mathbf{K}$. Suppose $M_0 \leq_{\mathbf{K}} M$ and $p \in \mathbf{gS}(M)$. We say $p \perp$ -does not fork over M_0 if for any/all a, N such that $\mathbf{gtp}(a/M, N) = p$, we have that $a \perp_{M_0}^N M$.

In some sense, the existence of a nice independence relation means our AEC has nice structure. Conversely, nice structural assumptions can be used to produce nice independence relations.

Background - Independence relations continued

We normally want our independence relations to satisfy some collections of the properties of forking in first order stable theories. A few examples:

Definition

Let $\perp$ be an independence relation on an AC $\mathbf{K}$. We say

1. $\perp$ satisfies *extension* if for every $M \leq_{\mathbf{K}} N$ and every $p \in \mathbf{gS}(M)$, there exists $q \in \mathbf{gS}(N)$ such that $p \subseteq q$ and $q \perp$ -does not fork over M .
2. $\perp$ satisfies *uniqueness* if for every $M \leq_{\mathbf{K}} N$ and type $p \in \mathbf{gS}(M)$, if $q_1, q_2 \in \mathbf{gS}(N)$ extend p and $\perp$ -do not fork over M , then $q_1 = q_2$.
3. for κ a regular cardinal, $\perp$ satisfies $(\geq \kappa)$ -*local character* if for every $\leq_{\mathbf{K}}$ -increasing chain $\langle M_i : i \leq \delta \rangle$ in $\mathbf{K}$ such that $\text{cf}(\delta) \geq \kappa$, where $\bigcup_{i < \delta} M_i = M_\delta$, and for every $p \in \mathbf{gS}(M_\delta)$, there exists $i < \delta$ such that $p \perp$ -does not fork over M_i .

Note that all the above hold if $\mathbf{K} = (\text{Mod}(T), \preceq)$ for a stable first order theory T and $\perp$ is first order non-forking.

Background - Classification theory

Abstract elementary classes provide a setting in which we can attempt a classification theory similar to that of first order theories. Some notions generalise easily - e.g. $\mathbf{K}$ is λ -stable if and only if for all $M \in \mathbf{K}_\lambda$, $|\text{gS}(M)| \leq \lambda$, and $\mathbf{K}$ is categorical in λ if and only if there is a unique model in $\mathbf{K}$ of size λ , up to isomorphism.

However, some notions do not lift immediately (e.g. NIP), partly because formulas are less naturally useful in this setting. And other notions have multiple equivalent definitions in first order that may not immediately coalesce in this more general framework (e.g. superstability).

Background - Test questions

One might hope that, like Morley's categoricity theorem, a tough problem might help us develop machinery in this framework to understand the more general picture. Probably the most important open problem in AECs is Shelah's categoricity conjecture:

Question (Shelah's categoricity conjecture (open))

For each $\lambda \geq \aleph_0$ and every AEC $\mathbf{K}$ with $LS(\mathbf{K}) \leq \lambda$, if $\mathbf{K}$ is categorical some $\mu \geq \beth_{(2^\lambda)^+}$, then $\mathbf{K}$ is categorical in all $\mu \geq \beth_{(2^\lambda)^+}$.

In the case $\lambda = \aleph_0$, for $k \in \omega$, under set theoretic assumptions there is an AEC $\mathbf{K}$ categorical in $\aleph_0, \dots, \aleph_{k-1}$, but not $\aleph_k$, so (unlike the first order case) $\mu_\lambda = \lambda^+$ is impossible.

Background - Progress

At this point, several approximations of the conjecture have been proven, including [GV06], [Vas18], [Vas17b], [Vas19], [SV24]. These all require some extra structure on the AEC or set theoretic assumptions (usually consistent with ZFC).

As hoped these have led to important discoveries in AECs, such as well behaved independence relations [Vas17a], [BG17], applicability of Ehrenfeucht–Mostowski models [She09], and the new notion of *tameness*.

Definition (Grossberg, VanDieren)

Let $\mathbf{K}$ be an AEC, $\mu \geq \text{LS}(\mathbf{K})$.

We say that $\mathbf{K}$ is μ -tame if for all $M \in \mathbf{K}$ and $p, q \in \text{gS}(M)$, if $p \neq q$, then there exists $M_0 \leq_{\mathbf{K}} M$ where $\|M_0\| \leq \mu$ and $p \restriction M_0 \neq q \restriction M_0$.

Many natural AECs satisfy some amount of tameness, and assuming existence of κ strongly compact, all $\mathbf{K}$ with $\text{LS}(\mathbf{K}) \leq \kappa$ are $(< \kappa)$ -tame [Bon14]. It is one of the assumptions of Grossberg and VanDieren's 'upward' categoricity transfer [GV06].

Background - Other areas of exploration

Some recent(ish) topics explored in AECs:

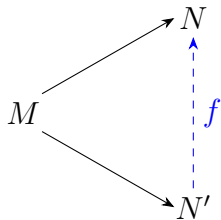
1. Axiomatisations of AECs [Leu23], [SV25]
2. Development of independence notions [LRV], [Vas17a], [BG17]
3. Generalisations to μ -AECs and μ -accessible categories [LRV]
4. Explorations of dividing lines in particular AECs [Maz21], [Maz25], [MR26]
5. Ultraproducts in AECs [Bon17]
6. Developing a notion of NIP for AECs [Yan25]
7. Equivalence of notions of superstability under tameness [GV17]
8. Behaviour of limit models in strictly stable AECs [Maz21], [Maz25], [MR26], [BV24], [the rest of this talk...]

Background - Universal extensions

Definition

Let $\mathbf{K}$ be an AEC with $\lambda \geq \text{LS}(\mathbf{K})$.

For $M, N \in \mathbf{K}_\lambda$, N is *universal* over M if $M \leq_{\mathbf{K}} N$ and whenever $N' \in \mathbf{K}_\lambda$ and $M \leq_{\mathbf{K}} N'$, there exists a $\mathbf{K}$ -embedding $f : N' \rightarrow N$ fixing M . We write $M \leq_{\mathbf{K}}^u N$.



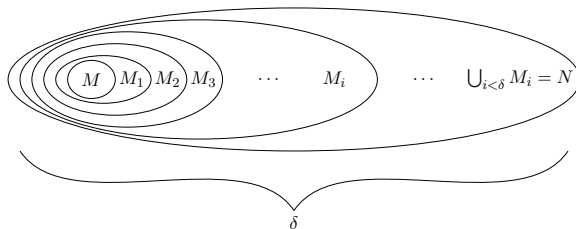
Intuitively, this says N contains a copy of every superstructure N' of M of size λ .

Background - Limit models

Definition

Let $\mathbf{K}$ be an AEC where $\lambda \geq \text{LS}(\mathbf{K})$ and $\delta < \lambda^+$ is a limit ordinal.

$N \in \mathbf{K}$ is a (λ, δ) -limit model over M if there exists a $\leq_{\mathbf{K}}^u$ -increasing sequence $\langle M_i : i < \delta \rangle$ in $\mathbf{K}_\lambda$ where $M_0 = M$ and $N = \bigcup_{i < \delta} M_i$.



Fact (Existence of limit models)

If $\mathbf{K}$ is λ -stable and $\mathbf{K}_\lambda$ has AP, then for any $M \in \mathbf{K}_\lambda$, there exists $N \in \mathbf{K}_\lambda$ universal over M . And for any $\delta < \lambda^+$ limit, there is a (λ, δ) -limit N over M .

Superstable - History of limit models

Proving that all limit models (over M) are isomorphic (over M) in certain settings is an important step towards Shelah's categoricity conjecture. Shelah and Villaveces drew up the blueprint of a proof under strongish assumptions [SV99], but there were errors in the proof.

Over the course of several papers, VanDieren corrected their proof and broadened the hypotheses [Van16]. Vasey developed this further and simplified elements of the proof. He proves that in all λ -superstable and λ -symmetric AECs, *all* limit models are isomorphic (we'll get to what this means soon).

All limit models in superstable AECs are saturated, so note that (in the superstable setting) this gives us a *strengthening* of the result that all saturated models are isomorphic. We will see this extends to the strictly stable setting also.

Background - Limit models

Fact (Uniqueness up to cofinality)

Suppose $\mathbf{K}$ is a λ -stable AEC and $\mathbf{K}_\lambda$ has AP, $\delta_1, \delta_2 < \lambda^+$ are limits with $\text{cf}(\delta_1) = \text{cf}(\delta_2)$, M is a (λ, δ_1) -limit model over M_0 , and N is a (λ, δ_2) -limit over M_0 . Then $M \cong_{M_0} N$.

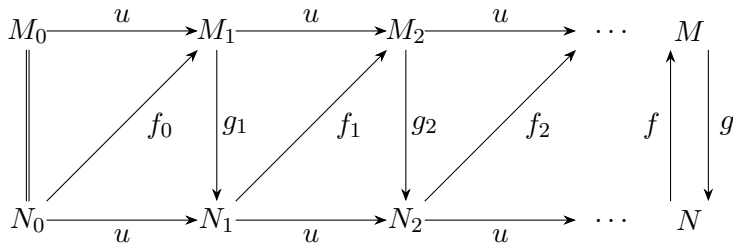
Moreover, if $\mathbf{K}_\lambda$ has JEP, M is a (λ, δ_1) -limit model and N a (λ, δ_2) -limit model, then $M \cong N$.

Proof: Let $\delta = \text{cf}(\delta_1)$. We may assume without loss of generality that $\delta_1 = \delta = \delta_2$. To see this, suppose $\langle M'_i : i < \delta_1 \rangle$ is a $\leq_{\mathbf{K}}^u$ -increasing sequence of models in $\mathbf{K}_\lambda$ with $M'_0 = M_0$ and $\bigcup_{i < \delta_1} M'_i = M$.

Then taking $\langle \alpha_i : i < \delta \rangle$ $<$ -increasing and cofinal in δ_1 with $\alpha_0 = 0$, we get that $\langle M'_{\alpha_i} : i < \delta \rangle$ witnesses that M is a (λ, δ) -limit model over M_0 . A symmetrical argument works for N , δ_2 .

Background - Uniqueness up to cofinality continued

Fix $\langle M_i : i < \delta \rangle$ and $\langle N_i : i < \delta \rangle$ both $\leq_{\mathbf{K}}^u$ -increasing continuous sequences in $\mathbf{K}_\lambda$ with $M_0 = N_0$ and $\bigcup_{i < \delta} M_i = M$ and $\bigcup_{i < \delta} N_i = N$. Then the following diagram can be built by a back and forth argument using that the sequences are $\leq_{\mathbf{K}}^u$ -increasing. For example, f_0 is just the identity, g_1 is obtained from $N_0 \leq_{\mathbf{K}}^u N_1$, f_1 from $M_1 \leq_{\mathbf{K}}^u M_2$, and so on. Take unions at limits.



Since everything commutes, f and g are inverses, and therefore isomorphisms. The 'moreover' part is obtained by a simple modification. □

Background - Limit models

With this in mind, we may as well focus on when δ is a regular cardinal, and we can talk about ‘the’ (λ, δ) -limit model (over M).

Definition

$\mathbf{K}_{(\lambda, \geq \kappa)}$ is the subclass of all (λ, δ) -limit models where $\delta \in [\kappa, \lambda^+)$ is regular.

Superstable - All limits are isomorphic

Theorem (Vasey)

Let $\mathbf{K}$ be a λ -superstable AEC with λ -symmetry for some $\lambda \geq \text{LS}(\mathbf{K})$. Then if $M, N_1, N_2 \in \mathbf{K}_\lambda$, and N_1, N_2 are limit models over M , then $N_1 \cong_M N_2$.

Moreover, all λ -limit models are isomorphic.

The λ -superstable and λ -symmetric assumptions for us basically say that

1. $\mathbf{K}_\lambda$ has AP, JEP, NMM
2. $\mathbf{K}$ is λ -stable (that is, for all $M \in \mathbf{K}_\lambda$, $|\text{gS}(M)| \leq \lambda$)
3. A particular independence relation called λ -non-splitting satisfies some of the nice properties forking in stable first order theories does - continuity, (universal) local character, and symmetry.

Note this is all 'local' to λ .

Outside of this superstable setting, limit models were largely unexplored.

Superstable - λ -non-splitting

λ -non-splitting is defined similarly to splitting in first order model theory, but replacing some finite tuples with λ -size models.

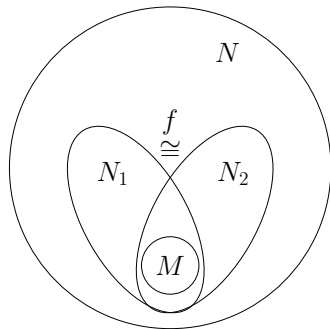
Definition

Let $\mathbf{K}$ be an AEC with $\lambda \geq \text{LS}(\mathbf{K})$. Suppose $M, N \in \mathbf{K}_\lambda$ with $M \leq_{\mathbf{K}} N$ and $p \in \mathbf{gS}(N)$.

We say p λ -splits over M if there exist $N_1, N_2 \in \mathbf{K}_\lambda$ where $M \leq_{\mathbf{K}} N_l \leq_{\mathbf{K}} N$ for $l = 1, 2$, and there is an isomorphism $f : N_1 \cong N_2$ such that $f(p \restriction M) \neq p \restriction N_2$.

Define $\perp_{\text{split}}$ by $a \perp_{\text{split}}^M N$ if and only if $\mathbf{gtp}(a/N, N')$ does not

λ -split over M . This is an independence relation, which we call λ -non-splitting.



Superstable - Unpacking the assumptions

The λ -superstable and λ -symmetric assumptions are quite strong, for a few reasons:

- Essentially we are assuming that a particular notion of independence (λ -non-splitting) is well behaved. Can we assume these for a more general notion of independence?
- In the first order setting, superstability is equivalent to saying we have full local character, i.e. $\kappa_r(T) = \aleph_0$. Can we weaken superstability to being (possibly strictly) stable, by assuming only $(\geq \kappa)$ -local character?

Beyond superstable - Existing result

An existing result in the strictly stable setting:

Theorem (Boney, VanDieren)

Let $\mathbf{K}$ be an AEC stable in $\lambda \geq \text{LS}(\mathbf{K})$ with AP, JEP, and NMM in $\mathbf{K}_\lambda$. Suppose $\kappa \leq \theta < \lambda^+$ are regular, and that $\perp_{\lambda\text{-split}}$ satisfies $(\geq \kappa)$ -local character, universal continuity, and (λ, θ) -symmetry.

If $M, N_1, N_2 \in K_\lambda$ where N_l is a (λ, δ_l) -limit over M and $\text{cf}(\delta_l) \geq \kappa$ for $l = 1, 2$, then there is an isomorphism from N_1 to N_2 fixing M .

So we should expect 'long' limits to be isomorphic. But what happens for a general independence notion, and what happens for limit models 'shorter' than κ ?

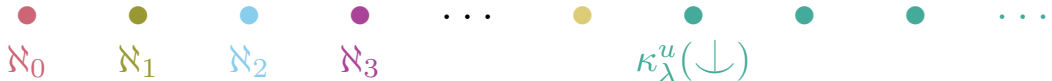
Strictly stable AECs - Main result

Theorem (B., Mazari-Armida)

Suppose $\mathbf{K}$ is an AEC stable in $\lambda \geq \text{LS}(\mathbf{K})$, with amalgamation, joint embedding, and no maximal models, and $\mathbf{K}$ is $\aleph_0$ -tame. Suppose $\perp$ is an independence relation on $\mathbf{K}_\lambda$ that satisfies uniqueness, extension, (universal) continuity, non-forking amalgamation, and $(\geq \kappa)$ -universal local character in some regular $\kappa < \lambda^+$. Let $\kappa_\lambda^u(\perp)$ be the least such κ .

Suppose $N_1, N_2, M \in \mathbf{K}_\lambda$, and $\delta_1 < \delta_2 < \lambda^+$ are regular cardinals, and N_l is a (λ, δ_l) -limit model (over M) for $l = 1, 2$. Then $N_1 \cong N_2$ (fixing M) if and only if $\delta_1 \geq \kappa_\lambda^u(\perp)$.

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Strictly stable AECs - Long limits are isomorphic

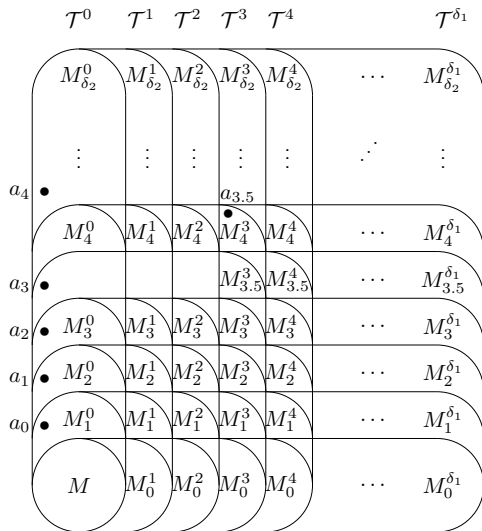
To prove the theorem, it is enough to find a single limit model M' which is both a (λ, δ_1) and (λ, δ_2) -limit model over M , as then by the earlier fact we have that

$$N_1 \underset{M}{\cong} M' \underset{M}{\cong} N_2$$

as desired.

But how do we construct M' ?

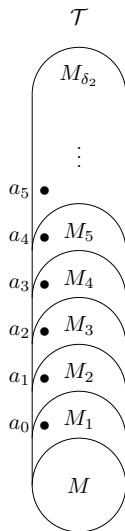
Strictly stable AECs - Long limits are isomorphic



To prove long limit models are isomorphic, we build a matrix of models, similar to [Vas19]. The method involves developing a theory of *towers*.

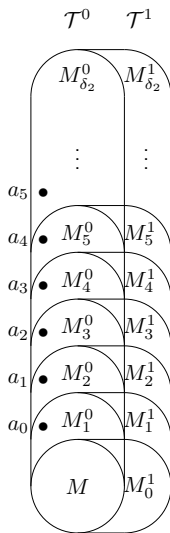
Compared to previous results, our method allows the use of general independence relations, possibly with $\kappa_\lambda^u(\perp) > \aleph_0$, and possibly defined only on $(\lambda, \geq \kappa)$ -limit models. $\aleph_0$ -tameness is not needed for this argument.

Strictly stable AECs - Long limits are isomorphic



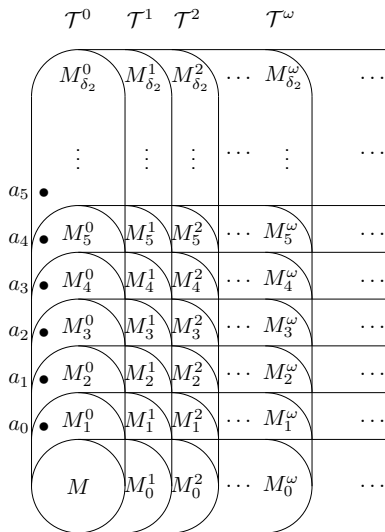
A tower $\mathcal{T}$ is a $\leq_{\mathbf{K}}$ -increasing sequence of $(\lambda, \geq \kappa_{\lambda}^u(\perp))$ -limit models M_i with named constants $a_i \in M_{i+1} \setminus M_i$. It is not hard to see towers of any length $< \lambda^+$ exist.

Strictly stable AECs - Long limits are isomorphic



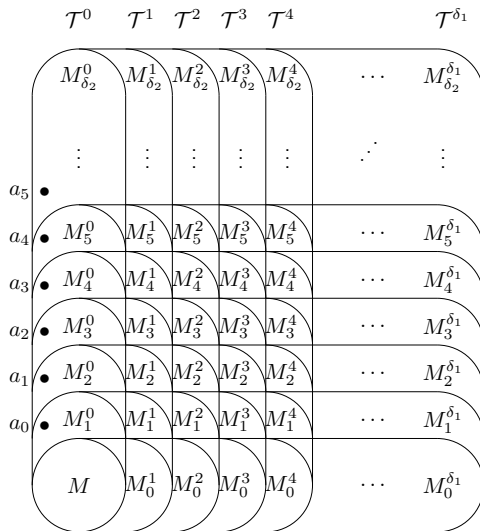
We define an ordering on towers $\mathcal{T} \triangleleft \mathcal{T}'$. One condition is that models in the same row must be universal extensions of previous towers. Also, we demand that the types of the $a_i \downarrow$ do not fork over models in the smaller tower. You can show that such extensions exist.

Strictly stable AECs - Long limits are isomorphic



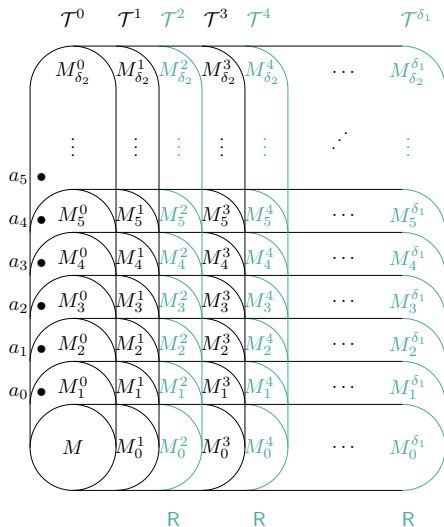
Unions of low cofinality towers may not be towers. So we also need to be able to extend arbitrary $\triangleleft$ -sequences of towers.

Strictly stable AECs - Long limits are isomorphic



To make the top row continuous, we need to ensure we can take unions of cofinality $\geq \kappa_\lambda^u(\downarrow)$ chains of towers.

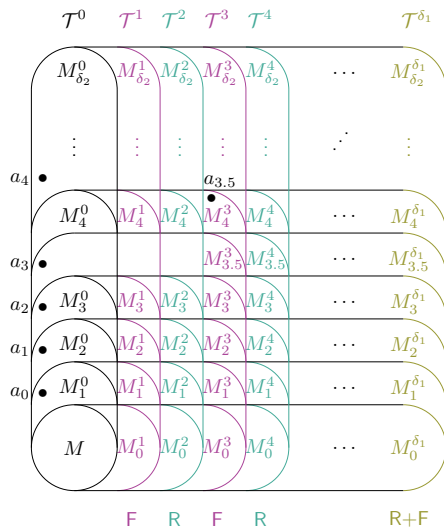
Strictly stable AECs - Long limits are isomorphic



Now we must make the final column universal and continuous at δ_2 . Since being universal and continuous are not preserved by unions, we use stronger conditions.

A tower $\mathcal{T}$ is *reduced* if the intersections of models in $\mathcal{T}$ with models in any tower $\mathcal{T}' \triangleright \mathcal{T}$ are as small as possible. Reduced extensions always exist, unions of reduced towers are reduced, and reduced towers are continuous at high cofinality limits.

Strictly stable AECs - Long limits are isomorphic



Finally we introduce *full towers*, which muddy the picture a little. We have to allow for adding new rows into our matrix as we go along. Brushing the details under the rug, full towers essentially make sure we insert enough types between the 'main' rows of the matrix to make the main rows universal.

We do this in a way that ensures full extensions always exist and are preserved by unions. This makes use of the independence conditions on the towers.

Strictly stable AECs - Short limits are non-isomorphic

We prove the contrapositive: that if two limit models of different cofinalities are isomorphic, then both cofinalities are 'high'. So suppose $\delta_1 < \delta_2 < \lambda^+$ are regular and the (λ, δ_1) -limit model and (λ, δ_2) -limit model are isomorphic. We will sketch why $\delta_1 \geq \kappa_\lambda^u(\perp)$.

Step 1: $\perp$ is 'close' to a relation called λ -non-splitting. This is enough to guarantee $\kappa_\lambda^u(\perp) = \kappa_\lambda^u(\perp_{\lambda\text{-split}})$.

Step 2: The union of any $\leq_K^u$ -increasing sequence $\langle M_i : i < \delta_1 \rangle$ is δ_1^+ -saturated.

Step 3: We have δ_1 -local character of non-splitting, so $\delta_1 \geq \kappa_\lambda^u(\perp_{\lambda\text{-split}})$. *This is where the $\aleph_0$ -tameness assumption is used.*

Step 1 and Step 3 combined give the conclusion.

Strictly stable AECs - Proof of step 3

Proof of step 3: Suppose for contradiction that $\delta_1 < \kappa_\lambda^u(\downarrow_{\lambda\text{-split}})$. Then there is a continuous $\leq_{\mathbf{K}}^u$ -increasing sequence $\langle M_i : i \leq \delta_1 \rangle$ in $\mathbf{K}_\lambda$ and some $p \in \mathbf{gS}(M_{\delta_1})$ such that p λ -splits over M_i for each $i < \delta_1$.

So there exist $S_i, T_i \in \mathbf{K}$ such that $M_i \leq_{\mathbf{K}} S_i, T_i \leq_{\mathbf{K}} M_{\delta_1}$ and $f_i : S_i \cong_{M_i} T_i$ such that $f_i(p \restriction S_i) \neq p \restriction T_i$ for all $i < \delta_1$.

By $\aleph_0$ -tameness, there exist countable sets $A_i \subseteq S_i$ and $B_i \subseteq T_i$ such that $f_i[A_i] = B_i$ and $f_i(p \restriction A_i) \neq p \restriction B_i$.

Let $C = \bigcup_{i < \delta_1} (A_i \cup B_i)$. Since $|C| \leq \delta_1 < \delta_1^+$, $p \restriction C$ is realised in M_{δ_1} . Let $d \in M_{\delta_1}$ realise $p \restriction C$. Fix $i < \delta_1$ such that $d \in M_i$.

But then

$$\mathbf{gtp}(c/B_i, M_{\delta_1}) = \mathbf{gtp}(f_i(c)/B_i, M_{\delta_1}) = f_i(p \restriction A_i) \neq p \restriction B_i = \mathbf{gtp}(c/B_i, M_{\delta_1}),$$

a contradiction!



Strictly stable AECs - Global version

If we assume we have a relation satisfying our conditions at all λ , the cardinal $\kappa_\lambda^u(\perp)$ stabilises for high enough stable λ to a value $\chi(\perp, \mathbf{K}, \leq_K^u)$. This gives the following:

Theorem (B., Mazari-Armida)

Assume $\perp$ is defined on $\mathbf{K}$ and has all the properties from before, with $\kappa \leq \text{LS}(\mathbf{K})$. Let $\lambda \geq \beth_{(2^{\text{LS}(\mathbf{K})})^+}$ such that $\mathbf{K}$ is stable in λ .

Suppose $\delta_1, \delta_2 < \lambda^+$ with $\text{cf}(\delta_1) < \text{cf}(\delta_2)$. Then for any $N_1, N_2, M \in \mathbf{K}_\lambda$ where N_l is a (λ, δ_l) -limit model over M for $l = 1, 2$,

$$N_1 \text{ is isomorphic to } N_2 \text{ over } M \iff \text{cf}(\delta_1) \geq \chi(\perp, \mathbf{K}, \leq_K^u)$$

Applications - First order stable theories

Given first order complete stable theory T , non-forking is a relations satisfying our hypotheses in $\mathbf{K} = (\text{Mod}(T), \preceq)$ with $\kappa_\lambda^u(\perp) = \kappa_r(T) \leq |T|^+$ for all stable $\lambda \geq |L(T)| + \aleph_0$. Thus we have:

Corollary

Let T be a first order complete stable theory. Then for every stable $\lambda \geq |T|$, and any $N_1, N_2, M \in \mathbf{K}_\lambda$ where N_l is a (λ, δ_l) -limit model over M for $l = 1, 2$ where $\text{cf}(\delta_1) < \text{cf}(\delta_2)$,

$$N_1 \text{ is isomorphic to } N_2 \text{ over } M \iff \text{cf}(\delta_1) \geq \kappa_r(T)$$

Though the $\Leftarrow$ implication was known, the $\Rightarrow$ implication was largely unexplored.

I found an alternative and much shorter proof of this theorem [Bea25b, Corollary 3.3], specific to this first order case, based on [GVV16, Theorem 7] (shortens the overall proof from 19 pages down to 2). Along the way, we prove that (λ, δ) -limit models are δ -saturated but not δ^+ -saturated for $\delta < \kappa_r(T)$.

Applications - Tame AECs

In fact, in nice μ -tame AECs, Vasey showed in [Vas16] the existence of an independence relation with many nice properties. With a little work, combining this with our 'long limit' result gives the uniqueness of $(\lambda, \geq \mu^+)$ -limit models for all stable $\lambda \geq \mu^+$.

Corollary

Let $\mathbf{K}$ be an AEC, stable in $\mu \geq \text{LS}(\mathbf{K})$ and stable also in $\lambda \geq \mu^+$, where $\mathbf{K}$ has AP, NMM, λ -JEP, and μ -tameness. Assume universal continuity of μ -non-splitting. Suppose also that $\mathbf{K}$ has μ -symmetry on $(\lambda, \geq \mu^+)$ -limit models.

Let $\delta_1, \delta_2 < \lambda^+$ be limit ordinals where $\mu^+ \leq \text{cf}(\delta_1), \text{cf}(\delta_2)$. If $M, N_1, N_2 \in K_\lambda$ where N_l is a (λ, δ_l) -limit over M for $l = 1, 2$, then there is an isomorphism from N_1 to N_2 fixing M .

Note we only require nice structural properties of the AEC - no independence relation is assumed to exist.

Applications - Algebraic AECs

There are several examples of algebraic AECs satisfying the assumptions of our theorem, but let's focus on one.

Let $\mathbf{K}^{R\text{-mod}}$ be the AEC of modules over a fixed ring R with $\subseteq$. In this case, $\chi(\perp, \mathbf{K}^{R\text{-mod}}, \leq_{\mathbf{K}}^u) = \gamma_r(R)$. In fact, considering when these are $\aleph_0$, we have

Corollary (Originally [Maz21])

The following are equivalent:

1. *R is Noetherian*
2. *for any $\lambda \geq \beth_{(2^{|R|+\aleph_0})^+}$, all λ -limit models in $\mathbf{K}^{R\text{-mod}}$ are isomorphic*

Weaker independence - Existing results

Recall the following 'long limits are isomorphic' result outside our setting:

Theorem (Boney, VanDieren)

Let $\mathbf{K}$ be an AEC stable in $\lambda \geq \text{LS}(\mathbf{K})$ with AP, JEP, and NMM in $\mathbf{K}_\lambda$, where $\kappa_\lambda^u(\downarrow_{\lambda\text{-split}}) < \lambda^+$, and $\downarrow_{\lambda\text{-split}}$ satisfies universal continuity and (λ, θ) -symmetry for some regular $\theta \in [\kappa_\lambda^u(\downarrow_{\lambda\text{-split}}), \lambda^+)$.

If $M, N_1, N_2 \in K_\lambda$ where N_l is a (λ, δ_l) -limit over M and $\text{cf}(\delta_l) \geq \kappa_\lambda^u(\downarrow_{\lambda\text{-split}})$ for $l = 1, 2$, then there is an isomorphism from N_1 to N_2 fixing M .

Weaker independence - the assumptions

λ -non-splitting is only known to satisfy weakened forms of uniqueness and existence, and the assumed form of symmetry is weakened also. Intuitively, 'universal wiggle room' is needed between the model of the type, and the model it is non-forking over. So our long limit result does not cover this case.

For example, $\perp$ has *weak uniqueness* if whenever $M_0 \leq_{\mathbf{K}}^u M \leq_{\mathbf{K}} N$ and $p \in \mathbf{gS}(M)$, if $q_1, q_2 \in \mathbf{gS}(N)$, extend p and $\perp$ -do not fork over M_0 , then $q_1 = q_2$.

Weak Indep. - Most general 'long limits are isomorphic' result with AP

The following is (to my knowledge) the most general 'long limits are isomorphic' result assuming AP:

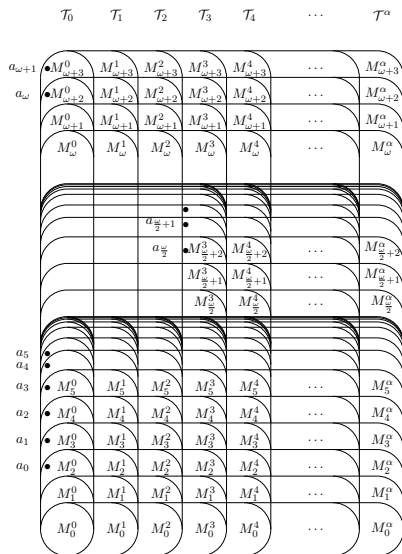
Theorem (B.)

Let $\mathbf{K}$ be an AEC stable in λ , where $\mathbf{K}_\lambda$ has AP. Let $\perp$ be an independence relation on $\mathbf{K}_{(\lambda, \geq \kappa)}$ satisfying weak uniqueness, weak extension, universal continuity in $\mathbf{K}_\lambda$, $(\geq \kappa)$ -local character, and (λ, θ) -weak non-forking amalgamation in some regular $\theta \in [\kappa, \lambda^+)$.*

If $M, N_1, N_2 \in K_\lambda$ where N_l is a (λ, δ_l) -limit over M where $\text{cf}(\delta_l) \geq \kappa$ for $l = 1, 2$, then there is an isomorphism from N_1 to N_2 fixing M .

Here universal continuity* in $\mathbf{K}_\lambda$ and (λ, θ) -weak non-forking amalgamations are appropriate weakenings of universal continuity and (λ, θ) -symmetry, so this generalises VanDieren and Boney's result.

Weaker independence - Problems and solutions



The new setting poses several challenges:

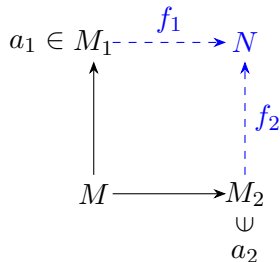
1. To apply the weak forms of uniqueness, extension, and transitivity, we need 'universal wiggle room'
2. In particular, the tower ordering may not be transitive!

To account for these, we modify the definition of tower, and instead of considering single towers, consider *brilliant chains of towers*, and *reduced chains*. With a lot of work, the proof goes through.

DAP - Disjoint non-forking amalgamation

Definition

We say that $\perp$ on $\mathbf{K}$ has *disjoint non-forking amalgamation* if for every $M_1, M_2 \in \mathbf{K}_{\parallel M \parallel}$ such that $M \leq_{\mathbf{K}} M_l$ and $a_l \in M_l$ for $l = 1, 2$, there exist $N \in \mathbf{K}_{\parallel M \parallel}$ and $f_l : M_l \rightarrow N$ fixing M such that $\text{gtp}(f_l(a_l)/f_{3-l}[M_{3-l}], M_3) \perp$ -does not fork over M , and $f_1[M_1] \cap f_2[M_2] = M$.



$$f_1(a_1) \underset{M}{\overset{N}{\perp}} f_2[M_2]$$

$$f_2(a_2) \underset{M}{\overset{N}{\perp}} f_1[M_1]$$

$$f_1[M_1] \cap f_2[M_2] = M$$

DAP - Disjoint non-forking amalgamation

Theorem (B.)

Let $\mathbf{K}$ be an AEC stable in λ , where $\mathbf{K}_\lambda$ has AP. Let $\perp$ be an independence relation on $\mathbf{K}_{(\lambda, \geq \kappa)}$ satisfying uniqueness, extension, non-forking amalgamation, universal continuity in $\mathbf{K}_\lambda$, and $(\geq \kappa)$ -local character. Then $\perp$ satisfies disjoint non-forking amalgamation.*

In particular, every $M \in \mathbf{K}_{(\lambda, \geq \kappa)}$ is a disjoint amalgamation base in $\mathbf{K}_\lambda$.

In fact, every $M \in \mathbf{K}_{(\lambda, \geq \kappa)}$ is a disjoint amalgamation base if we only assume weak uniqueness, weak extension, and (λ, θ) -weak non-forking amalgamation for some regular $\theta \in [\kappa, \lambda^+)$.

DAP - The construction

$$\begin{array}{ccc}
 \mathcal{T}^1 & \triangleleft_{(\lambda, \geq \kappa)} & \mathcal{T}^* \\
 M_1 = M_{\alpha,0} & \longrightarrow & M_{\alpha,1} \\
 \vdots & & \vdots \\
 \uparrow & & \uparrow \\
 M_{2,0} & \longrightarrow & M_{2,1} \\
 \uparrow & & \uparrow \\
 a_1 \in M_{1,0} & \longrightarrow & M_{1,1} \\
 \uparrow & & \uparrow \\
 M_0 = M_{0,0} & \longrightarrow & M_{0,1} \quad 'f[\mathcal{T}^2]' \\
 \parallel & & \parallel \wr f \\
 M_0 & \longrightarrow & M_2 \quad \mathcal{T}^2 \\
 & & \cup \\
 & & a_2
 \end{array}$$

There are three main components to the proof:

1. Show we can form a reduced tower with top model limit over the bottom
2. Show we can make a small reduced tower where $\langle M_0, M_{1,0} \rangle^\wedge \langle a_1 \rangle$, and glue this to the bottom of the reduced tower
3. Show we can 'amalgamate' this to the tower $\langle M_0, M_2 \rangle^\wedge \langle a_2 \rangle$, respecting the independence.

Since the first tower is reduced, $M_1 \cap f[M_2] = M_0$, so the four corners of the picture make the desired amalgam.

If we sacrifice the 'non-forking' requirement, the proof is easier and works with the 'weak' assumptions.

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Thanks for listening!